

VR Avatars for Older Adults

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1 Abstract

With the introduction of new technologies to the masses, older adults are often left behind. Virtual Reality (VR) applications, which require avatar creation, mainly ignore aged representations. In order to make VR avatar customisation more inclusive for older adults, this thesis investigates the preferences of older adults regarding VR avatars in the existing studies, and presents a prototype to demonstrate a VR avatar creator tailored for older adults. Reflecting on the research and the current state of the art, we discuss the features and adjustments needed for better inclusion of older adults in VR avatar customisation, such as the possibility of creating a realistically looking aged body with specific posture, body type, gray hair and corresponding fashion choices. By implementing these features in the VR avatar creator prototype, this work helps to pave the way for further research and proper development of VR applications for older adults.

2 Introduction

Virtual reality (VR) technologies are employed to simulate environments and create immersive experiences for users. Nowadays, VR is applied across various fields, such as training, education, and entertainment [19]. Educational VR provides effective learning experiences, enabling users to explore objects and events that are otherwise inaccessible, such as the solar system or historical sites [19].

In VR, users typically experience the “presence” - an illusion of being within the simulated environment, *“in spite of the fact that you know for sure that you are not actually there”* [34]. This experience is maintained through an immersive VR system - a technology, that “delivers the ability to perceive through natural sensorimotor contingencies” [34], and is enhanced by the use of avatars, which serve to represent a body, that is possessed by the user in that system. Customization of these avatars allows users to tailor their digital presence to better suit their preferences. Popular applications and games like VRChat [28] and Skyrim VR[25] offer extensive avatar customization tools. However, modern avatar creators predominantly focus on youth and beauty, often neglecting more diverse representations [9]. It looks like current systems do not sufficiently meet the needs of older adults, thereby a focused research might be needed in this area. For instance, many existing systems lack options for representing an aged body, focusing instead on youthful and idealized body types [9] with limited customization features such as wrinkles, grey hair, or age-related body shapes. Furthermore, these platforms often feature modern or trendy clothing and accessories, neglecting styles that may be more familiar or preferred by older adults. Additionally, it seems like supportive devices such as canes, walkers, or wheelchairs are also rarely available, what might be limiting the accuracy and relatability of avatars for older adults. One of the research goals in human-computer interaction

(HCI) is to understand and compensate for age-related changes with digital technologies, provide older adults with necessary tools or improvements, and help them live as independently as possible [1]. VR has the potential to address all three objectives. It can help older adults maintain social ties with distant family and friends, providing a sense of inclusion and accomplishment [33]. Additionally, avatars can offer anonymity and safety in interactions, encouraging older adults to engage in activities like exercise [4]. This opens up the question, whether older adults would hesitate to engage with new technologies. Studies indicate that 360° video interventions are perceived by older adults as both safe and enjoyable [18]. These findings challenge the opinion that older adults are universally unenthusiastic about new technologies, highlighting the potential of VR to positively impact their wellbeing. Moreover, VR can benefit aged care residents and those affected by dementia or Alzheimer’s [12], providing them with activities that might help overcome social isolation, feelings of loneliness, creating calming experiences to address anxiety and agitation [5].

Since VR could have a positive effect on older adults in general, and avatars play a crucial role in VR, appropriate avatar customization should be developed to maximize the benefits of the technology for older adults. This is the primary focus of this study. Avatars are crucial in VR because they serve as the user’s representation in the virtual world, greatly influencing their sense of presence and immersion. The appearance and behavior of an avatar can affect how users perceive themselves and interact with the virtual environment and other users. For older adults, having avatars that meet their personal preferences can enhance their comfort and engagement with VR applications. This can lead to improved social interaction, a greater sense of inclusion, and potentially significant psychological benefits. Addressing this topic is essential for making VR a more inclusive and beneficial tool for older adults. This study tries to approach the problem, guided by the following research questions:

- RQ1 What kind of thoughts and considerations regarding customization options of VR avatars can be found in existing literature?
- RQ2 What would an interactive VR avatar creation tool tailored for older adults look like?

Thereby the work have been done in acquiring the knowledge for the possible customization options from various articles and existing tools, and creating a VR prototype based on that accumulated knowledge as an example to demonstrate how a VR Avatar creator might be developed with older adults in mind.

3 Related Work

Although not much research has been conducted specifically on the intersection of VR, avatars, and older adults, relevant studies exist across overlapping fields. In this section, we provide a general

understanding of VR, the concepts of presence and immersion, and dive into the sense of embodiment, especially in the context of older adults.

3.1 Virtual Reality Concepts

A virtual environment (VE) refers to a computer-generated three-dimensional space that can be explored and interacted with using various interfaces [14]. VEs are widely utilized in domains such as computer gaming, film production, and simulation training, where the world is typically displayed on a monitor and controlled using input devices like keyboards or controllers [33]. In such environments, users often interact through a character or a body representing them within the space. VR technology, particularly immersive virtual environments (IVEs), distinguishes itself from other VEs by not only representing a space but engaging the user's sensory and motor systems to create a deeply immersive experience [34]. Unlike traditional interfaces, users interact with a virtual world through simulated sensory substitution, often involving individual displays for each eye, volumetric sound, and tracking of the body parts location. Here, natural sensorimotor contingencies, or how users perceive and act within a space, play a crucial role [34]. In this context, the user's agent in the environment, or their virtual body and an avatar, is becoming perceptible. When analyzing this perception, the concept of the Sense of Embodiment (SoE) is becoming essential [17]. Referring to the user's experience of owning and controlling a body within an immersive environment, it includes three interrelated components: Sense of Self-Location, Sense of Body Ownership, and Sense of Agency: (i) The Sense of Self-Location refers to the perceived position of the self within a virtual body and a virtual world, with a common finding that an altered position can lead to something like out-of-body experiences [13], (ii) A presence of the Sense of Body Ownership implies that the user perceives a virtual body as their own. The famous Rubber Hand Illusion experiment demonstrates how a foreign object shaped like a part of the body - a rubber hand - can be intuitively perceived as that part itself [8]. The same effect can be achieved through virtual body parts. (iii) The Sense of Agency describes the feeling of control over one's virtual body, and unlike the Sense of Body Ownership, can be rather actively perceived, for example through a deliberate hand movement [16], and might be extended to the Tool Embodiment, where users perceive the objects they control and interact with - such as mobility aids - as extensions of their bodies [17]. This principle might have an impact on designing avatars for older adults, especially when representing walkers or wheelchairs. Although SoE is difficult to measure objectively, researchers agree that deficits in any one of these components can lead to a reduced sense of embodiment, and that all three should be considered when trying to achieve a better perception of a virtual body [17].

Proper immersion in a virtual body leads to interesting experiences. Because the avatar is perceived as the user's own body, the user may, intentionally or unintentionally, start to care about it's wellbeing. For example, in the study by Argelaguet et al. [3] participants were tasked with moving their virtual hands to avoid simulated obstacles. Despite the absence of physical risk towards their actual bodies, users exhibited avoidance behaviors. Furthermore, immersion in a virtual body affects the behaviour of the users

[38]. A study by Banakou et al. [6] reports, that participants embodied in an avatar resembling Einstein exhibited improved cognitive performance, aligning their behavior with the perceived attributes of their virtual body. This phenomenon, termed the Proteus Effect [38], demonstrates how users adapt their actions to match the expectations associated with their avatars. Moreover, further studies have extended this research to non-human avatars. For instance, users embodied as animals reported heightened empathy for wildlife [2], while those embodied as plants described a sense of tranquility and interconnectedness with nature [35]. These findings suggest that avatar design can greatly influence psychological outcomes and user engagement. Additionally research indicates the importance of relatability between users and their avatars. Greater similarity between a user's appearance or desires and their avatar correlates with higher satisfaction and immersion [7]. However, this relationship might become complicated by the uncanny valley effect, where avatars that are overly realistic and at the same time imperfect become unsettling, with factors such as facial expression synchronization and motion realism impacting the realism and attractiveness of avatars [11].

3.2 Virtual Reality and Older Adults

A common theme in research on technology use by older adults is the growing willingness of this demographic to engage with digital tools and platforms. For instance, online gaming—traditionally dominated by younger users—is becoming increasingly popular among older adults [9]. However, while technologies adoption rates are rising, challenges remain in making them accessible and beneficial to this group [33]. Research often focuses on helping older adults adjust to technological advances, compensating for any physical or cognitive limitations that may hinder effectiveness of use.

A specific discipline in research, Human-Computer Interaction (HCI) focuses on human-centered design, pointing up importance of creating technologies that help with the needs, preferences and abilities of users. A book by Sayago et al. [1] highlights three overarching goals of HCI research in the context of older adults: (i) Understanding and compensating for age-related changes in functional abilities to ensure older adults can effectively interact with digital technologies, (ii) Enhancing the quality of life for older adults by either introducing technologies specifically designed for them or simplifying and adapting existing tools to better suit their needs, (iii) Supporting independent living and aging-in-place through assistive technologies. Lining up with these principles, VR technology might have a potential to benefit older adults in various domains. For example, unlike traditional video calls or chats, VR offers immersive social experiences through embodiment, enabling older adults to participate in shared activities in a more engaging way [1]. Moreover, VR has been shown to support cognitive abilities, enhance reaction times, and improve physical coordination [37].

While VR presents promising opportunities, accessibility remains a critical barrier for older adults. Many applications are designed with younger users in mind, with interaction methods such as VR controllers optimized for youthful agility. Older adults often face challenges with control schemes, physical discomfort, or motion sickness [37]. A study by Wu et al. [37] addresses these concerns, questioning the usability of VR interfaces and control

mechanisms. For instance, older participants in that study exhibited lower accuracy in pointing tasks or reported heightened physical fatigue during prolonged sessions. Efforts to improve VR accessibility for older adults include redesigning interaction techniques and providing more intuitive training and interfaces [37]. These solutions work towards reduction of discomfort and making VR a viable tool for this demographic.

A study by Baker et al. [4] researchers have sought to understand older adults' perceptions of VR and its potential as a tool for social interaction. Findings reveal that while some older adults are hesitant about the technology due to its novelty, many see its value in fostering connections and providing meaningful engagement. Participants noted that VR could help overcome barriers like geographical separation, creating spaces for shared activities such as games or sports, storytelling, or even virtual travel. Regarding specifically avatar creation, it becomes crucial to understand how older adults negotiate virtual representations of themselves. The work of Carrasco et al. [9] serves as a foundational work in this area, exploring older adults' preferences for avatars. Participants in the study used MakeHuman [22] software to create their avatars, with researchers analyzing the resulting designs to identify recurring themes and preferences. Participants' choices were categorized into four main types of avatars: (1) Actual Avatars, closely resembling the participants' real-life appearance, often intended for activities where authenticity or familiarity was important; (2) Vibrant Avatars, idealized versions of oneself with enhanced physical attributes, reflecting youthful energy or aspirational traits; (3) Other Avatars, representing entirely different identities—such as another person, animal, or fictional character—used for exploration or experimentation; and (4) Companion Avatars, created to provide companionship or emotional support, indicating the participants' desires for connection. This typology suggests that avatar preferences may be context-dependent, with the choice of an avatar varying based on the intended activity, audience, or desired self-perception. For example, older adults might prefer realistic avatars for professional or family interactions but opt for vibrant or other avatars when engaging in gaming or exploration. In addition to Carrasco et al.'s work, other studies have delved into factors such as realism and cultural stereotypes in avatar design. Preferences for realism vary, with some users appreciating highly detailed avatars while others feel more comfortable with stylized or abstract representations [29]. Cultural and generational influences may also shape how older adults perceive and design avatars, from clothing choices to face and body types [11]. These findings highlight the complexity of avatar design for older adults, drawing attention to the need for customizable tools that accommodate diverse preferences. For this research, Carrasco et al.'s insights [9] provide a starting point for understanding older adults' avatar needs and inform the development of VR avatar creation tools tailored specifically for this demographic.

3.3 Existing avatar creators

Several avatar creators were explored to identify strengths, limitations, and potential for adaptation in creating a VR avatar creator with older adults in mind.

Makehuman [22], a tool mentioned in the previous section is an open-source application designed for creating characters with customizable body shapes. It features an extensive set of sliders to control specific body features, allowing users to morph between various body types. For example, gender is represented as a continuum, with users able to transition fluidly between male and female forms. The application supports the customization of hair, clothing, and poses, and it allows the extension of settings for advanced customization. It also includes options to export models in various file formats, such as FBX or OBJ. A key feature for this research is the tool's ability to represent aging bodies. It includes a slider for age, which adjusts posture, adds wrinkles, and applies gravitational effects to body mass. Custom skin textures, including aged ones, can also be uploaded. Above mentioned study by Carrasco et al. [9] utilized MakeHuman to allow older adults to create avatars, demonstrating that some participants, who wished to achieve realistic representations of aging bodies, could do so, as seen in the figure 1.

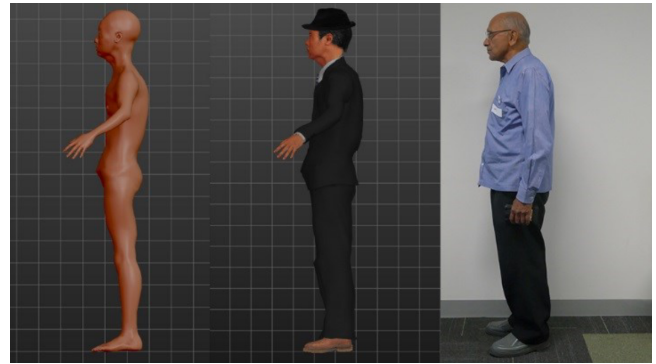
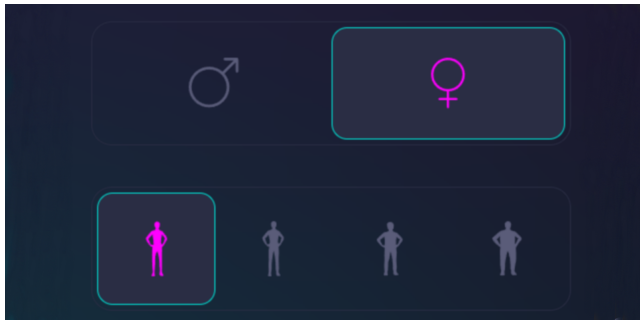


Figure 1: Avatar created by a participant of a study by Carrasco et al. [9], using the tool Makehuman.

Ready Player Me [24] is a platform for creating avatars online, with the ability to deploy them into various applications. Its interface is simple and user-friendly, with customization options such as face shape and outfits. However, it is limited in terms of body types (figure 2a), offering only variations in body weight. While it includes numerous options for hairstyles and clothing, these largely seem to be mainstream preferences, with no options for creating aged avatars (figure 2b). This limits its utility for older adults.

Avaturn [20] specializes in creating realistic avatars based on uploaded selfies. Its customization options are comparable to those of Ready Player Me, with limited body types and numerous choices for hairstyles and clothing. However, Avaturn also allows for exaggerated body proportions, such as oversized heads or elongated arms, primarily for entertainment purposes. While it provides a unique feature for creating personalized head and face, its body customization remains minimal, and there is little focus on aged representations.

MetaHuman [23], powered by Unreal Engine, is a state-of-the-art solution for creating realistic characters. It features an intuitive tool for blending facial features from multiple presets and allows users to adjust shapes directly using control points. While it offers



(a) Body presets



(b) Available clothes

Figure 2: Customisation settings in Ready Player Me [24]

some presets for creating older-looking faces (figure 3a), its options for aged body proportions and postures are limited (figure 3b).

The diversity of existing avatar creation tools might complicate the research. Avatar creators designed for games, social platforms, or multipurpose applications each require distinct design considerations. The presence of VR introduces yet another variable, with only a small subset of the tested avatar creators operating in VR environments. In fact, many popular VR applications, such as VR-Chat [28], bypass the need for integrated avatar creation, instead allowing users to import external 3D models.

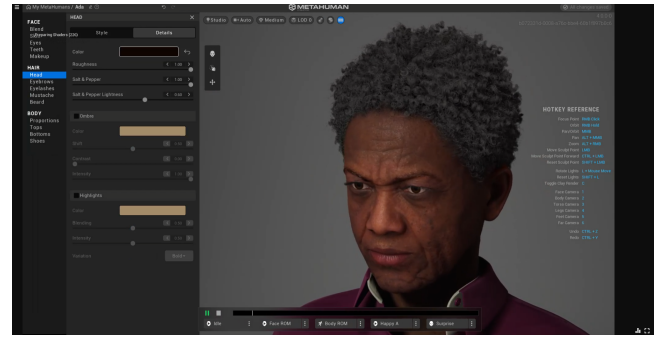
In summary, while existing avatar creators demonstrate impressive capabilities in customization, most of them fall short in addressing some of the specific needs and preferences of older adults. These might include the ability to represent aged bodies types.

4 Methods

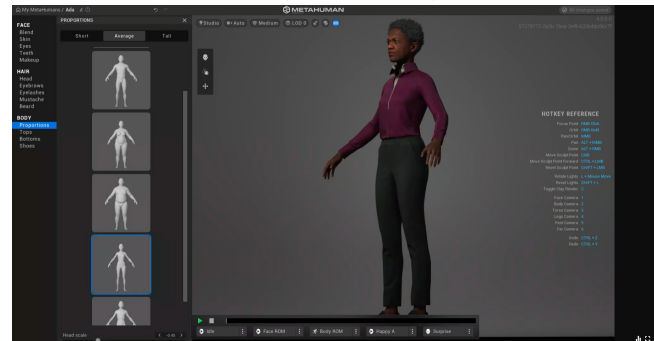
The focus of this research lies in the creation of a practical example of a VR avatar creator specifically designed for older adults. In order to establish a pool of most important and relatable features, an extensive review of various existing studies have been conducted to gather insights into older adults' preferences and the technical requirements for an avatar creation tool. This process also involved examining challenges faced during the prototype's development and implementation.

4.1 Accumulating preferences of older adults

Several studies have documented the preferences of older adults in relation to avatar creation and virtual environments.



(a) Creating an old looking face



(b) Available body presets

Figure 3: Creating an older looking person in MetaHuman [23]

In this research, we employed a manual process to generate a list of preferences based on existing data from the studies, listed in the Table 1. Initially, all relevant information, including research notes, participant statements, feedback on applications used, and any other applicable details, were compiled. Following this, the items were grouped by related subjects to help organize the data. The next step involved identifying and extracting object-related information, searching for specific mentions within the data. For example, many articles emphasized the desire to have control over body musculature, the option to add a beard, or preferences for reduced hair on the head. This process led to the creation of a compiled sheet that showed the accumulated preferences, both general and specific, for avatars and avatar creators. These insights guided development of the VR avatar creator prototype.

4.2 Development of VR Avatar Creator Prototype

The VR Avatar Creator for older adults was developed using a combination of software tools. Unity [27] 2022.3.41f1 served as the foundational engine for the project, providing the framework for designing, building, and deploying VR applications. UMA [26] v2.11 was chosen as the base system for avatar customization, offering a robust set of standard options for creating and manipulating virtual characters. Blender [15] played a crucial role in editing and exporting 3D models and textures, while Krita [21] was used for

Article Title
Bridging the Generational Gap: Exploring How Virtual Reality Supports Remote Communication Between Grandparents and Grandchildren [36]
Comparing avatar game representation preferences across three age groups [31]
Age and Realism of Avatars in Simulated Augmented Reality: Experimental Evaluation of Anticipated User Experience [29]
Exploring the Design of Social VR Experiences with Older Adults [4]
Older Adults Designing Avatars for Socializing [10]
To be (me) or not to be? photorealistic avatars and older adults [30]
Negotiating stereotypes of older adults through avatars [9]
The Social Interaction Experiences of Older People in a 3D Virtual Environment [32]
Avatar: a virtual face for the elderly [11]

Table 1: Studies used as sources for research on older adults’ preferences in VR avatar customization

editing textures. Unity is an engine that provides tools for scripting, interface design, and VR integration. Unity’s capabilities were used to create a customizable and interactive environment, with VR functionality handled through its integrated VR toolkit. This allowed interaction and embodiment within the virtual space. The core of the avatar customization was built upon the Unity Multipurpose Avatar (UMA) framework. UMA enables dynamic generation and customization of humanoid models, starting from a provided base model. Customization was mostly achieved by manipulating the underlying rigged bones, which directly modified the model’s proportions and shape. This bone-driven modification system contributes in tailoring body features to user preferences. Additional textures and models, such as clothing and overlays, were integrated and managed through UMA, which supports the layering and management of multiple visual assets on the humanoid base. This modularity enables the addition of aged textures and clothing models, aligning with older adults’ preferences.

5 Results

5.1 Accumulated preferences of older adults

An analysis of existing literature provided insights into preferences of older adults for avatar customization. These preferences were manually selected based on subject they related to, and compiled into the Table 2, with specific examples illustrating each preference. The data reveals some preferences, which focus on specific avatar features, such as finger’s length or beard options, others are more abstract, addressing general customization of clothing or body types. The bottom sections general statements and avatar specifics are dedicated to broader statements, that occurred in the feedback from

participants about avatar creators, which they were using during respective studies.

5.2 VR Avatar Creator Prototype

The development of the VR Avatar Creator prototype focused on implementing basic features derived from the preferences accumulated in the previous section. According to these findings, different older adults preferred avatars that represented a variety of appearances, including different ages (from young, such as grandchildren, to their own), various races (to represent people of different ethnicities and cultures), diverse body types (from short and slender to tall and robust), and various genders, including nonbinary. This highlights the importance of offering vast customization possibilities, making the diversity of appearances a central goal of the prototype. These overarching characteristics, such as age, gender, or race, influence multiple specific bodily features. For instance, the position of the cheeks might be influenced by age but could also represent individual distinctions. While a top-down approach—where overarching characteristics directly determine all specific features—would be too complex, this prototype adopts a bottom-up approach. By adjusting individual features, users can create avatars that align with desired overarching characteristics, while maintaining unique personal details. To ensure these overarching characteristics can be achieved, the prototype provides ample customization options for individual features. For example, older-looking avatars can be created by adjusting skin textures, posture, muscle mass, and weights, as well as by modifying facial features such as the mandible, jawline, forehead, eyes, and cheeks. Hairstyles also play a significant role, allowing adjustments for hair density, graying, and baldness. Similarly, racial representation can be achieved using the same controls, including changes to skin color and facial structure. The prototype also implements customization options based on specific preferences identified in studies. These include a height adjustment setting to create avatars of varying stature, a beard option for users who desired facial hair, the ability to adjust fingers’ length (as noted in the study with a participant named Forjador[9]), settings to create older-looking skin textures (as noted by participant Gordon[9]), and controls to adjust the width of body parts such as arms to reflect muscular changes. Gender is the only overarching parameter directly adjustable in the prototype. Changing this setting influences the base model and available customization options. Currently provides only male and female genders. Further, since most of the studies emphasized the importance of the face, without more distinct directions, the face customisation options aim to provide diversity without being based on specific preferences. Control over facial emotions was not implemented for this prototype, but many studies generally expressed that some controls over emotions would be preferred. The same applies to outfits: while accumulated references include wishes for more clothing, but without specific proposals, the prototype only tried to expand on the standard wardrobe that comes with UMA, providing 13 additional wardrobe items. Recognizing the interest of some older adults in non-human avatars, the prototype also includes options for alternative representations, such as a car, a deer, and a chicken. Other considerations from the studies were also implemented. Female customization was designed to be as detailed as male customization.

Preference type	Subject	Example
Specific choice for a characteristic	Age	Forjador created a 30 years old avatar [9] Nik created a 78-year old avatar [9]
	Race	Mariam created an avatar with asian ethnicity [9] Klaus created an avatar with average race (30% Asian, African and Caucasian) [9]
	Gender	Amy created an "androgynous" avatar [9]
	Hight	Nik created a tall female character(238 cm) [9]
	Eyes	Gordon created avatar with blue green eye color [9] Red Dog created an avatar looking her grandson that has eye color he could "trust" [9]
	Hair	Participants suggested adding reduced hair of head [30][9][10]
	Beard	Participants mentioned that they would like to add a beard [9]
	Torso	Keki modified the depth and position of the avatar torso and hip depicting some similarity with his body shape [9]
	Hands	Forjador expressed a preference for long fingers [9]
	Skin	Gordon applied 'old skin' texture for his avatarGordon applied 'old skin' texture for his avatar [9]
	Muscles and Weight	Sally wanted an avatar with "good physique" [9]
		Red Dog created an avatar looking his grandson that was "well built" [9]
		Nik created an avatar with high muscle mass [9]
	Posture	Tony added a low level of 'pregnancy stomach' trying to simulate their body figure [9]
	Outfit	Keki modified the depth and position of the avatar torso and hip depicting some similarity with his body shape [9] "I chose to wear military uniform [in VR] because I always wore it when I was young." [36]
General customizability of a characteristic	Face	"The majority of the participants emphasized the facial features being the most important part of the avatar in assessing the avatars." [30]
	Emotions	Participants expressed need to convey emotions, specifically positive [9][30]
	Non-human	Participants allowed possibly of having a non-human representation for specific context [4][9][32]
General statements	Outfit	Nik was concerned she could not find suitable clothing for her avatar [9]
	Options for women	Female participants wanted to have more options for clothes or hairstyles [9][32]
Avatar creator specifics	Faithful representation	Some participants criticized that avatars looked younger than the age they had chosen [9]
	Mirror	Participant expressed a wish to see their real appearance in order to reconstruct it properly as avatar [9]
	User experience	Some participants wanted the controls to be simpler and less time-consuming [31]

Table 2: The compilation of preferences of older adults accumulated from multiple studies

Faithful aged representations were prioritized, avoiding deliberate alterations that might make avatars appear younger than intended by users.

UMA operates through a system of components, with the Dynamic Character Avatar (DCA) being central. This component connects to a human-representation prefab and manages various parameters, such as race, wardrobe recipes, shared parameters, and DNA values. UMA's standard content includes male and female base models, each with predefined settings, wardrobe recipes, and DNA configurations. These base models, along with their settings, are grouped into "races." Customization in UMA relies on DNA parameters, which define the morphing and structural adjustments of the character's body. DNA configurations are stored in Dynamic UMA DNA Assets and applied through Dynamic DNA Converter Controllers. These parameters allow for precise manipulation of skeletal and mesh transformations. For example, the length of fingers can be adjusted by modifying the X-scale of specific finger bones, while posture changes—an important feature for older adult representation—were introduced by altering the rotation of spine bones and repositioning the shoulders. Adjustments to skin tone and other visual features are managed through UMA's overlay and texture systems. UMA overlays, are used to dynamically map textures to specific areas of the character model. For example, applying older skin textures to a DCA can be achieved through UMA's overlay system, which uses the same UV layout as the base texture. A texture representing aged skin was created using a reference photo of an older male head scan from a publicly available website. This photo was edited to match the UV layout of the UMA base texture, as shown in the figure 4b ensuring proper alignment and applied as a skin overlay over the avatar's model, demonstrated by the figure 4d.

Changing avatars clothes is managed by UMA's wardrobe system, which organizes clothing and accessories into slots, such as Body, Hair, and Hands. Each slot is populated by slot recipes, which define the textures, meshes, and overlays associated with that category.

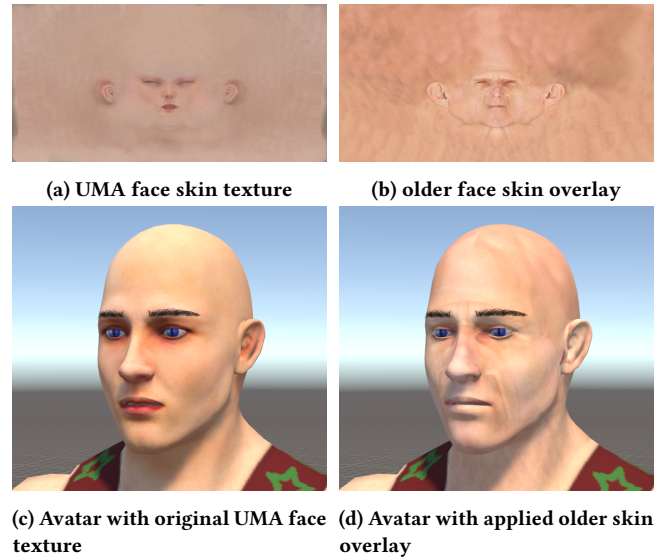


Figure 4: Applying skin texture customisation with UMA

Various slot recipes applied onto avatars can be seen in the figure 5. Slot recipes can be race-specific or shared across multiple races. Also, a specific use of slot recipes "No Head" and "No Hair" is applied to the virtual body, which is directly controlled by the user, hides internal head geometry, preventing rendering issues in VR.

Expanding the wardrobe system involved sourcing new 3D models and converting them into slot recipes. These models were then imported into Blender for adjustments. The clothing geometry needed to be adapted to fit the UMA race model precisely, ensuring that the body did not poke through the clothing and that the rig binding was correctly applied. After making these adjustments in Blender, the modified clothing was exported into Unity.



Figure 5: Customized characters with multiple wardrobe recipes applied

Within Unity, the UMA framework was used to create assets and slot recipes, allowing the clothing to be applied to Dynamic Character Avatars (DCAs), as shown in the figure 6.



(a) Clothes model edited in Blender (b) Clothes set up for an UMA character

Figure 6: Adjustment process of custom clothes models

Unity's original VR Template provided a framework for rendering and controlling the application with a VR setup. This template enabled the system to track the positions of the headset and controllers, translating user inputs into actions within the virtual world. These inputs formed the basis of the VR controller setup and provided functionality for pointer-based interactions and spatial UI design. The interface allows users to interact with sliders, toggles, and buttons using a mouse, VR controllers, or joysticks. Additionally, dynamic panels can be moved, scaled, and scrolled with VR controllers. The VR interface serves as the foundation for the Humanoid Avatar Creator, a system divided into five states: Skin, Body, Head, Hair, and Outfit. Each state populates a menu with specific controls relevant to the selected category. This design allows users to navigate and manipulate avatar features through direct VR interactions.

To enable virtual body movement, one of the UMA character was set up with an Inverse Kinematic (IK) Rig. Unity's built-in solutions, such as the Two Bone Constraint and Multiparent Constraint, were utilized for this purpose. These constraints allowed the rotation of controllers to be translated to the avatar's bones, while the head was fixed to the position of the VR headset.

To achieve a matching sense of embodiment (SoE), adjustments and offsets were implemented for the virtual body: since users have different physical heights, the system includes an option for VR controller adjustments. This was accomplished through T-pose calibration, where users are required to stand in a T-pose to measure their arm span and height. These measurements are used to ensure alignment between the real-world and virtual body. A reference T-pose of the virtual body is maintained by one of the DCA clones, which remains in a T-pose for calibration purposes, but invisible for the user. To account for differences between the real-world and virtual body dimensions, the system applies adjustments to the virtual body based on the reference and measured T-pose data. These adjustments allow the virtual body to reflect changes in height, limb length, or body proportions. For instance, longer arms result in a wider reach, while user in control over taller avatar experience the virtual world perceived smaller.

In order to give user possibility of seeing their customized virtual body from a perspective, a virtual mirror was created. A simple implementation involved a plane acting as a mirror surface, with an invisible camera placed behind it. This camera projects the user's position and movements onto that plane, creating the illusion of a reflection. A key challenge was to display the whole body, with the virtual body lacking head parts, since they are removed by "No Head" and "No Hair" wardrobe slot recipes. To address this, another DCA clone was created, which is complete with every body part, but visible only through the mirror's camera. This clone completely mirrors the movements of the original virtual body while including head parts, thus providing an accurate reflection.

In the resulting prototype, users can create avatars with varying levels of detail by navigating through a menu-driven interface. The figure 7 shows how it looks like within the application. The system provides menus for body customization, skin adjustments, facial features, hairstyles, and clothing selections. For example, the Body Menu allows users to modify the avatar's shape, posture, and body-weight, while the Skin Menu includes options for selecting older skin textures and adjusting skin tones. In the Head Menu, users can reshape facial features such as the jawline, nose, and eyes, and the Hair Menu provides a range of hairstyles with adjustable hair density and color, including graying options. The Outfit Menu adds flexibility by letting users mix and match different clothing items, such as tops and bottoms, to personalize their avatars further. In addition to humanoid avatars, the system offers a unique feature that allows users to embody non-humanoid entities, such as animals. For example, a user can select a deer as a character and experience a simplified embodiment of this virtual form. The system also emphasizes embodiment, providing users with the ability to control of a virtual body that may significantly differ from their real-world form. This is achieved by translating real-world movements into virtual inputs, enabling users to experience alternative body types. For example, avatars with exaggerated features, such as extended

arms, enable an expanded range of interaction. The virtual environment itself is customizable, with settings consisting of a living room, RPG-style village and an abstract plane with empty space.

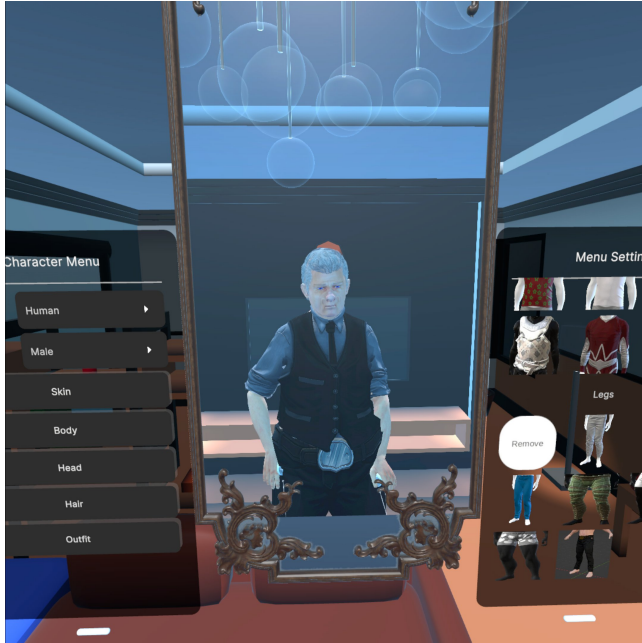


Figure 7: Screenshot from the VR Avatar Creator prototype, showcasing an older-looking male avatar designed using the customization options available in the system.

6 Discussion

With this work, we aggregated findings about VR avatars for older adults based on existing literature and demonstrated an exemplary implementation of these insights within a VR scene.

In designing the prototype, an important assumption was that a successful VR avatar creator for older adults must allow users to create avatars that closely reflect their desired representations. For some, this involves crafting an avatar that mirrors their aged body, complete with realistic proportions, skin texture, and other features. The inclusion of aging body characteristics is a key aspect of the prototype, as previous research from related works has indicated that some older adults prefer avatars that resemble their current selves, despite societal pressures favoring youthfulness. For instance, male participants in studies, such as by Carrasco et al. [9], have shown an interest in creating older-looking avatars, demonstrating a desire for authenticity in their life stage. To address this, the prototype offers detailed options for customizing aged features, including body proportions and posture, older skin textures, facial features, and hair with age-related density and color changes. Unlike many existing platforms, such as Ready Player Me [24], which provide limited and homogenized body type options, this prototype offers finer control. Users can independently adjust posture, weight, and muscle mass, capturing the diverse physical variations that come with aging. While not as powerful

as the professional options in MakeHuman [22], the prototype is a reasonable compromise, providing enough complexity for meaningful representation without overwhelming users with technical details. The inclusion of aged skin textures is vital, as it is nearly impossible to accurately portray an aged body without them. Some platforms offer automated tools to generate realistic textures from selfies, such as Avaturn [20], but this prototype focuses on manual control over individual facial features, such as the chin, cheeks, and nose. While this approach requires more effort, it does give the user the ability to create a highly personalized representation. Another area where the prototype is quite innovative is hair customization. Users can select hairstyles and colors, as found in nearly all avatar creators, but this prototype also introduces the ability to adjust hair density. Although this feature currently relies on symbolic adjustments to hair transparency, it does a good job of conveying the visual impression of aging hair. However, further refinement would increase its realism. Clothing customization, a crucial aspect of self-representation, is included, but with limitations. While older adults often emphasize the importance of having sufficient choice of clothing for avatars, the scope of this work restricted the addition of culturally or age-specific outfits. Developing a broader range of clothing options that align with older adults' preferences and cultural contexts might be an area for future research. In addition to visual customization, the prototype uses the immersive capabilities of VR to enhance the user experience. Embodiment is partially implemented to improve the sense of "being" within the avatar. It allows users experience their avatars from a first-person perspective, including adjustments to height and arm span. For example, users can create avatars with exaggerated physical traits, such as extremely long arms, and experience how these changes affect their interaction with the environment. This embodiment feature not only enhances immersion, but also introduces a fun element by allowing users to experiment with fantastical representations. This fits with findings from studies such as by Baker et al. [4] where older adults expressed a desire to use VR to relive physical experiences or experiment with abilities their bodies no longer allow. Context remains an underexplored topic, though. Research suggests that users might benefit from being given context for avatar creation, but the prototype offers only limited environments: a cosy living room, a fantasy RPG village, and an empty world. Expanding the range of contexts to include environments such as family gatherings or professional settings, could provide users with a clearer purpose for their avatars, potentially affecting their customisation choices.

Like many software projects, the prototype development process faced challenges. Initially, Ready Player Me [24] was considered as the basis for the avatar system, but the fact that it relied on server-side model generation meant that further adaptation was not possible. Instead, the project focused on the use of UMA. Because UMA's default models were designed for average, youthful body types, adapting UMA to create aged representations was still challenging. Limited documentation also meant that much of the implementation relied on trial and error, adding significant overhead to the development process. In spite of these problems and complications, the project has succeeded in producing a working prototype. The use of existing tools such as UMA and Unity accelerated development, but introduced accessibility limitations as

these platforms were not originally designed with older adults in mind. For example, standard VR controls and inputs in Unity may not meet the needs of older users, suggesting the need for tailored accessibility solutions for future solutions. Overall, the iterative and agile development process allowed us to focus on essential features, resulting in a prototype that demonstrates the potential of a VR avatar creator tailored for older adults.

6.1 Limitations

The scope of this thesis may be a little overwhelming, with a very wide range of topics. On the other hand, it justifies the exploratory and rather open-ended type of research that can benefit several disciplines at the same time.

One limitation lies in the reliability of the preferences and findings from the research. It should be considered that there may be additional studies relevant to this topic that were not reviewed or could not be found during the research process. Many of the studies reviewed, such as by Carrasco et al [9], were based on small samples of participants, raising questions about how the findings can be generalised. This limited number of participants makes it uncertain whether the preferences identified truly reflect the majority of older adults or are specific to a particular subgroup. This potential for error means that conclusions drawn from these sources should be treated with caution, keeping in mind the evolving nature of research in this area.

The usability of the prototype for older adults is another area of concern. Although the standard Unity VR template controls feel responsive, they may be too complex or unintuitive for older users, particularly those unfamiliar with VR technology. For example, older adults may find it difficult to interact with the application without seeing their real hands, or may struggle with abstract VR controls. The prototype's interface, which combines multiple menus with sliders and buttons, presents its own challenges. A minimalist design risks missing essential features, while an overly detailed interface could overwhelm users. Finding the right balance remains a challenge.

The current menu organisation, which sorts options alphabetically, may not be a user-friendly flow. A more intuitive approach would prioritise high-impact or logically sequenced options. For example, options with the most visible or significant effect on the avatar could appear first, improving the overall user experience.

Furthermore, the prototype is not without its technical problems. Known bugs include jittering of the avatar during customisation and delays when loading humanoid models. These problems arise from the interaction between UMA, Unity Animator and rigging systems. Fixing these issues would require deeper knowledge and further development. In addition, compatibility issues with clothing models result in occasional clipping between the avatar and the clothing, further harming the user experience. Further testing and refinement would be required to resolve these issues.

The lack of certain features also limits the effectiveness of the prototype. For example, the avatar does not perform walking or crouching animations, and its default 'hovering' pose is a noticeable problem. Users who sit during customisation will find their avatars clipping through the floor, which detracts from immersion. Adding a sitting mode or options for avatars using mobility aids such as

wheelchairs or walkers would improve realism as well as address the needs of older adults with mobility issues.

The prototype's rigging system for VR control is very simple. Elbow positioning and shoulder joints are stiff and unnatural, limiting the avatar's ability to accurately imitate real-world movements. This problem is particularly visible when users try to extend their arms fully forward. Further development could introduce a better rigging system to ensure that virtual limbs more closely match real-world movements.

Also while the prototype strives for inclusiveness, there are still significant flaws. For example, it does not currently support non-binary gender options, as implementing this feature would have required creating entirely new systems from scratch, which was beyond the resources of the project. Similarly, skin tone customisation is limited to a single slider, which does not take into account the complex variations in hue and saturation found in different skin types. Expanding this feature to include a wider range of tones and textures would improve inclusivity.

The aging skin texture, which is currently only applied to the face, does not apply to other visible body parts such as the hands. This omission can detract from overall realism, as hands are often visible in VR and are critical to immersion. A full body texture, with variations to reflect diversity, would significantly improve the user experience.

6.2 Outlook

This research provides a foundation for understanding and addressing the unique needs and preferences of older adults in the context of VR avatar creation. While we have implemented features aimed at increasing inclusiveness for this demographic, the actual added value of such a tool for older adults remains an open question. Further research is needed to evaluate the practical and emotional benefits that older adults might derive from engaging with this technology.

Getting older adults to engage with VR technology could be seen as a challenging task. Despite the many reported benefits of VR - such as stimulating social connectivity, promoting creativity, and providing options to remote traveling - older adults are often sceptical about adopting the new technology. This hesitation may be due to lack of familiarity, perceived complexity, or concerns about practicality and relevance to their daily lives. As such, the motivation for older adults to engage in avatar creation for digital activities, whether social or recreational, may not be immediately obvious to them. Nevertheless, the development of this prototype aims to demonstrate the potential for an avatar creator tailored to the preferences of older adults. By demonstrating the possibilities of customisation and embodiment in VR, this work hopes to inspire further efforts to make VR applications more inclusive. The prototype acts as an example of how user-centred design principles can be applied to address the often overlooked needs of older adults.

Still, a critical limitation of this work is that the prototype was not tested with older adults. Although interviews and qualitative testing were planned, the development process took more time than anticipated, leaving no opportunity for direct involvement with the target demographics. As a result, it is impossible to determine whether the research goals were actually met, or the extent to which

the proposed solutions meet the actual preferences and needs of older adults. Future studies would benefit from involving actual older adults in the evaluation process to gather valuable feedback and validate the approach.

While this project is not strictly tied to a specific online platform, game or social application, the lessons learned could inform the design of future tools in different contexts. Developers who consider the preferences and accessibility needs of older adults from the very beginning could significantly improve the appeal and usefulness of their applications. Ultimately, encouraging inclusivity in avatar creation and VR experiences could ensure that older adults are not left behind as virtual and augmented realities become part of modern life.

7 Conclusion

This thesis tried to investigate the preferences of older adults regarding VR avatars, using insights from existing literature to identify key customization features and interface considerations. Building on these findings, a VR avatar creation prototype was developed to explore how these preferences could be implemented practically.

The research revealed the diversity of older adults' preferences, such as flexibility in age, gender, race, body type and detailed customisation options. While the prototype was able to reflect many of these preferences, the lack of direct testing with older adults remains a critical limitation. Without feedback from the target population, the effectiveness of the prototype in achieving its intended goals might not be fully verified. Nevertheless, this work provides a solid foundation for future efforts to make VR more inclusive and accessible for older populations. It stresses the importance of tailoring technology to meet the needs of different user groups and demonstrates the potential for further innovation in creating meaningful and engaging VR experiences.

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Appendix: Asset References

Asset Description	Link
Mirror 3D Model	https://sketchfab.com/3d-models/mirror-9-mb-755d7092bb1049efa92d5aaa47e06ffd
Female Outfit 3D Model 1	https://www.cgtrader.com/free-3d-models/character/clothing/womens-outfit-e73d5258-04a1-4020-8964-408155825aab
Female Outfit 3D Model 2	https://www.cgtrader.com/free-3d-models/character/woman/woman-with-dress-free
Female Outfit 3D Model 3	https://www.turbosquid.com/3d-models/genesis-3-clothing-40-1098627
Bodyguards 3D Model	https://assetstore.unity.com/packages/3d/characters/humanoids/humans/bodyguards-31711
Animals 3D Models	https://assetstore.unity.com/packages/p/animals-free-low-poly-asset-pack-by-ithappy-260727
Surge for Unity	https://assetstore.unity.com/packages/tools/utilities/surge-107312
UniRx	https://assetstore.unity.com/packages/tools/integration/unirx-reactive-extensions-for-unity-17276
Interior 3D Model	https://free3d.com/3d-model/living-room-997877.html
Interior Materials	https://assetstore.unity.com/packages/3d/props/interior/house-interior-free-258782
RPG 3D Environment	https://assetstore.unity.com/packages/3d/environments/landscapes/rpg-poly-pack-lite-148410
Male Skin Texture	https://www.3d.sk/photos/show/id/0003-man-head-premade-texture-0003-33
Female Skin Texture	https://www.3d.sk/photos/show/id/emilia-rouco-head-premade-texture